Same Gene, Different Tissue: The Prognostic Weight of ERBB2

BIO392 - Bioinformatics of Sequence Variation

ERBB2 in Breast Cancer

- Focal amplification on 17q12
- Found in roughly 15 to 20% of invasive breast carcinomas
- Drives constitutive HER2 dimerization, RAS/MAPK and PI3K/AKT activation
- Before targeted therapy, HER2-positive breast cancers had shorter disease-free and overall survival

Tang et al., 2024

ERBB2 in Lung Adenocarcinoma

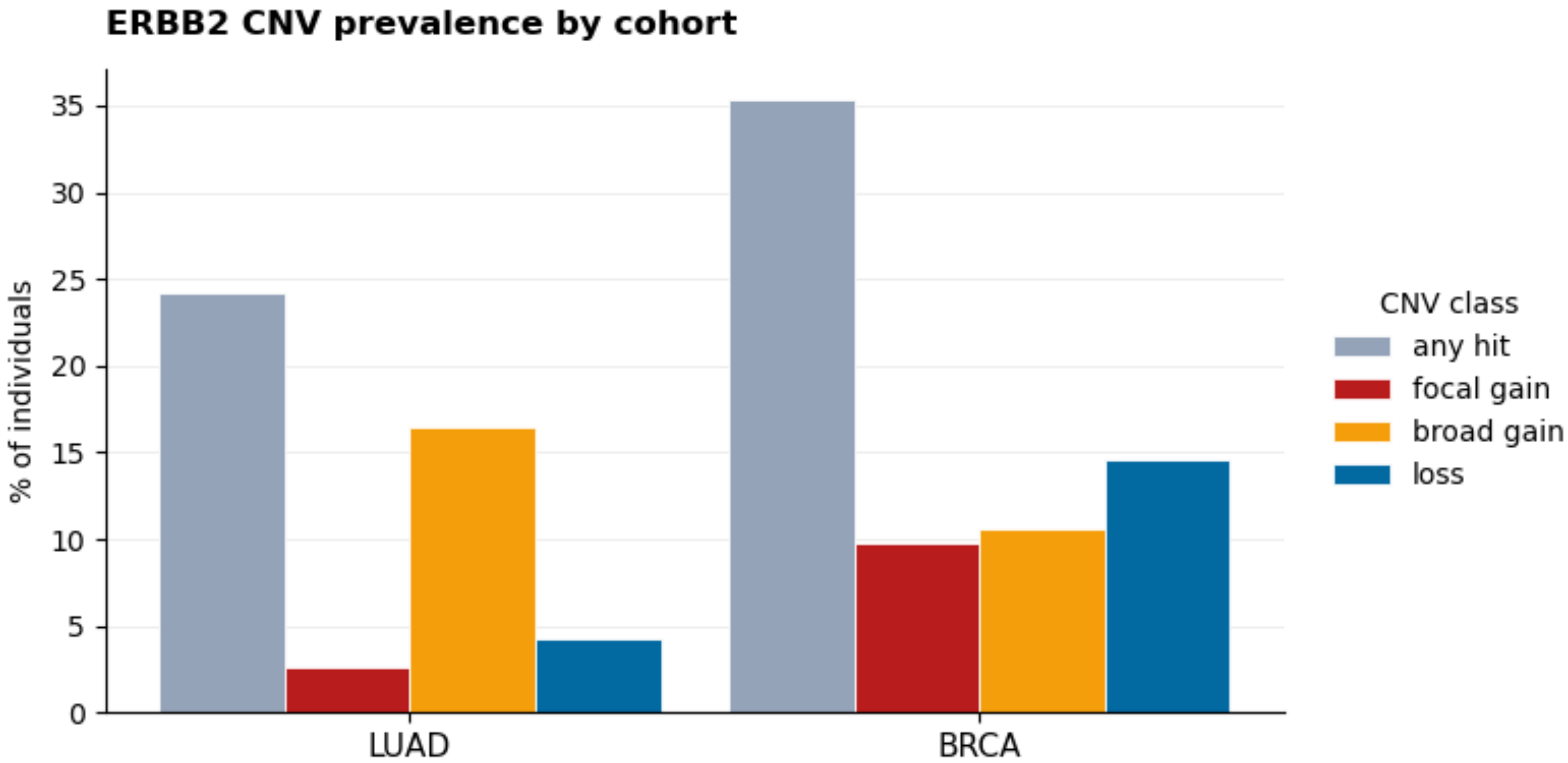
- ERBB2 amplification or activating mutation is reported in roughly 2 to 4% of NSCLC
- Predominantly in adenocarcinoma
- Often in non-smokers without EGFR or KRAS drivers
- Reported associations with poorer outcome in several LUAD cohorts

Ismail et al., 2025

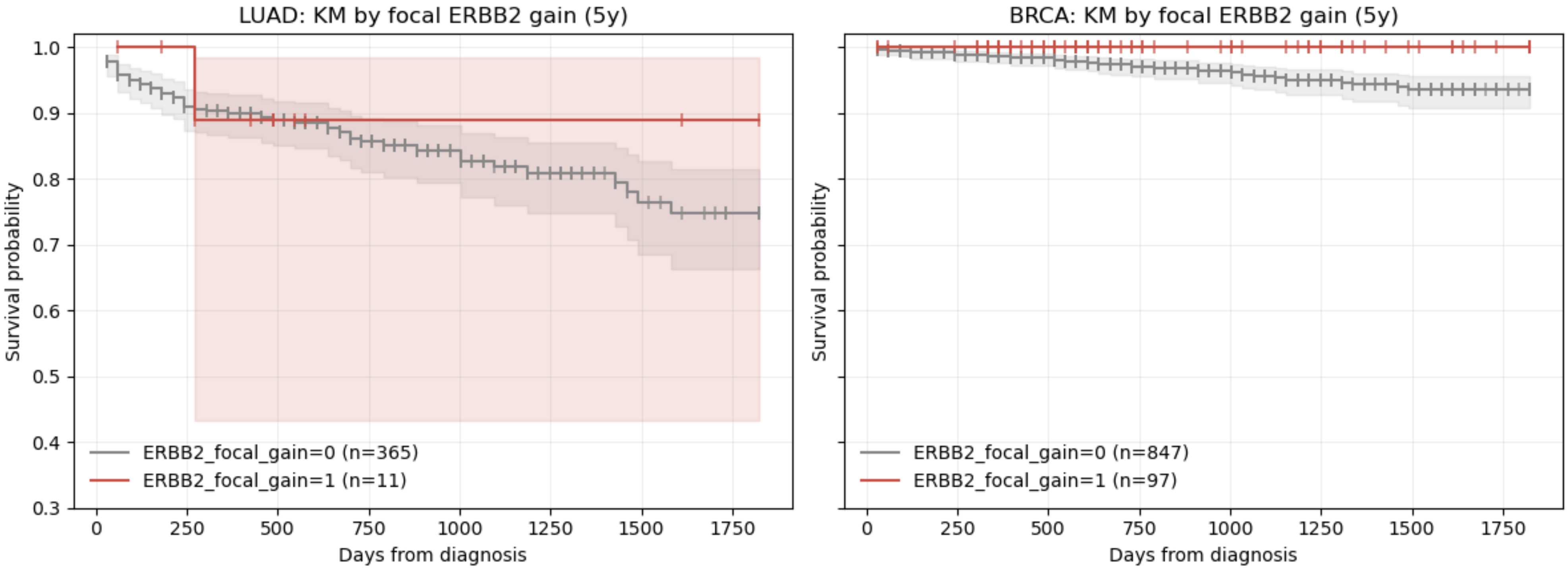
TCGA Cohorts

- Both cohorts are pulled from Progenetix
- LUAD n=376 (57 deaths, 15% event rate)
- BRCA n=944 (44 deaths, 5% event rate)

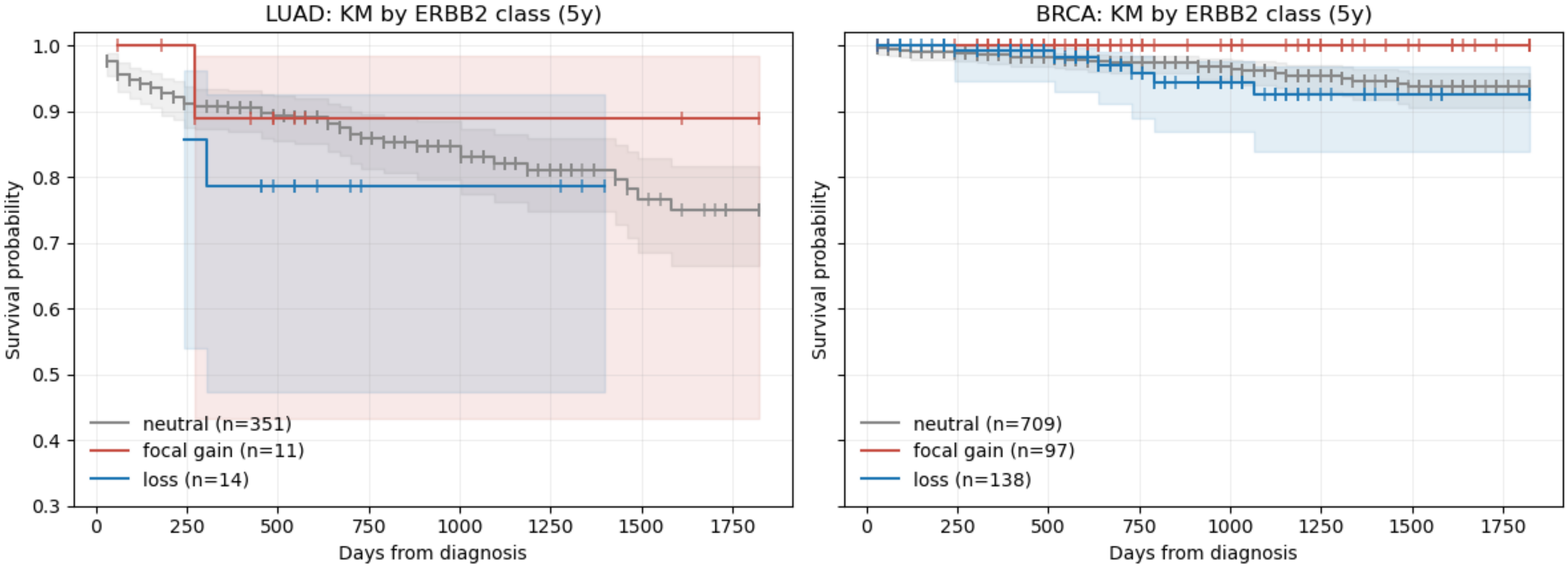
ERBB2 CNV Prevalence



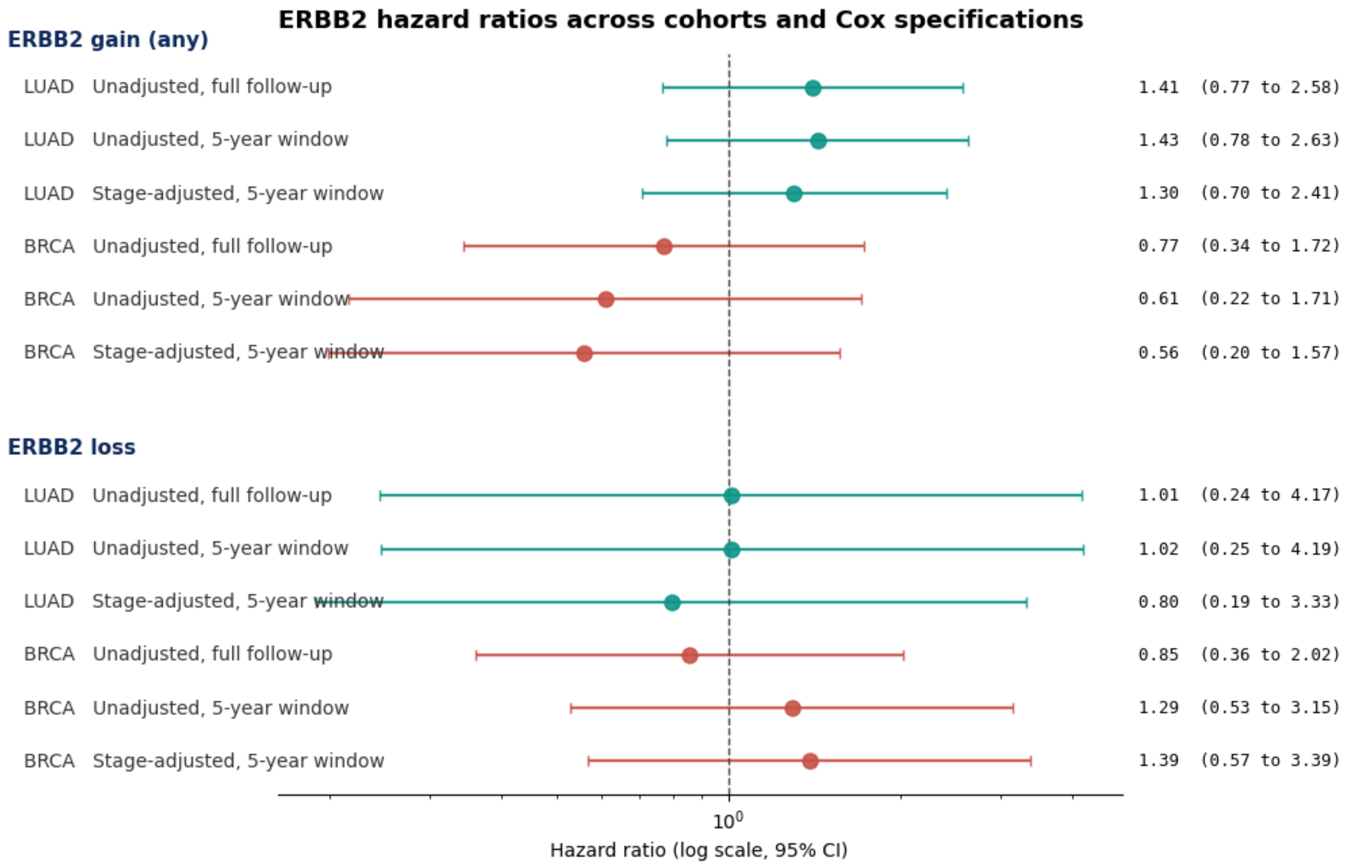
ERBB2 Gain Profiles



ERBB2 Gain vs. Loss



ERBB2 Hazard Ratios



Discussion

- Stage is the dominant prognostic variable
- Stage adjustment lifts the C-index from 0.52 to 0.62 in LUAD and from 0.53 to 0.69 in BRCA

Discussion

- In LUAD, ERBB2 gain trends toward worse survival (HR ~1.3 to 1.4)
- In BRCA, it trends toward better survival (HR ~0.3 to 0.6)
- Opposite of the ”HER2-positive equals worse outcome”
  - Maybe: Large-scale chromosomal instability, not focal ERBB2 loss
  - Maybe: Large 17q losses can drag in BRCA1

Caveats

- Study is not powered to formally detect the ERBB2 effect in BRCA
- Only 92 BRCA focal-gain patients and 32 total events in the 5-year window